Strategic Optimization of IoT Energy Management Systems: Aligning Dynamic Tariff Calculation Algorithms with Sri Lanka's May 2026 Electricity Tariff Revisions

Executive Summary and Macro-Contextual Framework

The global hospitality sector is currently undergoing a profound technological transformation, driven by the dual imperatives of environmental sustainability and stringent operational cost management. In the context of Sri Lanka, a nation heavily reliant on its tourism industry for foreign exchange and macroeconomic stability, small-to-medium-sized enterprises (SMEs) such as tourist villas, guesthouses, and homestays form the backbone of the accommodation sector.[1] However, these SME operators face a severe operational vulnerability: the unchecked consumption of electrical energy by guests, colloquially termed "behavioral waste".[1] Guests routinely leave high-load appliances—most notably air conditioning (HVAC) systems and water heaters—running while absent from their rooms.[1] Traditional mitigation strategies, such as key-card power switches, have proven highly susceptible to guest bypass mechanisms, while enterprise-grade Building Management Systems (BMS) remain prohibitively expensive for the local SME market.[1]

Compounding this technological deficiency is the extreme volatility of the Sri Lankan electricity market. On May 9, 2026, the Public Utilities Commission of Sri Lanka (PUCSL) approved a sweeping revision of electricity tariffs, fundamentally altering the financial landscape for commercial operators effective May 11, 2026.[2] Driven by surging global fuel prices and a projected revenue shortfall of approximately 38 billion LKR across the current and upcoming quarters, the National System Operator necessitated an aggressive cost-recovery strategy.[2] While a 15 billion LKR government subsidy shields 95% of domestic consumers (specifically those utilizing under 180 units per month), the commercial and high-consumption sectors face stark tariff escalations.[2] General Purpose (GP) consumer categories are now subjected to a rigorous 18% baseline increase across their energy and fixed demand charges.[8]

Crucially, in a strategic macroeconomic maneuver designed to protect the fragile but vital tourism sector, the PUCSL explicitly exempted small and medium-scale hotels operating under the specific H-1 and H-2 tariff categories from this 18% hike.[8] This regulatory bifurcation creates an absolute operational imperative for SME tourist accommodations: they must formally secure an H-2 classification to avoid punitive GP-2 rates, and they must deploy intelligent, context-aware Energy Management Systems (EMS) capable of dynamically navigating the Time-of-Use (ToU) structures native to the H-2 category.

The subsequent analysis provides an exhaustive architectural, economic, and programmatic blueprint for optimizing a Smart IoT-Based Energy Management System. By meticulously detailing the Dynamic Tariff Calculation Algorithm and aligning it with the realities of the May 2026 PUCSL tariff doctrine, this report establishes a definitive framework for achieving real-time Return on Investment (ROI) quantification and systemic grid demand reduction.

Macroeconomic Drivers of the May 2026 Tariff Revisions

To fully engineer a responsive cost-saving algorithm, the macroeconomic catalysts driving the May 2026 tariff revisions must be systematically understood. The Ceylon Electricity Board (CEB) operates within a highly sensitive thermal-hydro generation mix, rendering it exceptionally vulnerable to external economic shocks. In early 2026, shifting hydrological inflows and escalating international fossil fuel benchmarks severely impacted the generation dispatch cost profile.[10] The CEB identified a massive revenue deficit—initially projected at 15.8 billion LKR for the second quarter, which rapidly escalated to a systemic 38 billion LKR shortfall across broader forecasting horizons.[2]

The Subsidy and Sectoral Cross-Subsidization

In response to widespread public and political backlash regarding the inflationary pressures of electricity costs on the general populace, the Sri Lankan government intervened. A 15 billion LKR subsidy was injected into the grid ecosystem, designed to remain active until September 30, 2026.[2] This financial buffer effectively neutralized the proposed tariff increases for domestic consumers and religious institutions consuming fewer than 180 units monthly, shielding the most economically vulnerable segments of the population from immediate harm.[2]

However, utility economics dictates that unrecovered generation costs must be absorbed elsewhere to maintain the fiscal viability of the grid. Consequently, these deferred costs were aggressively shifted onto high-consumption and non-exempt commercial tiers. Large-scale industries, government institutions, and general commercial entities classified under the General Purpose (GP-2 and GP-3) categories were hit with the absolute maximum threshold of the revision—an immediate 18% increase across both volumetric energy charges and fixed monthly demand charges.[2]

The Hospitality Exemption: A Strategic Forex Buffer

The PUCSL recognized that subjecting the recovering tourism sector to an unexpected 18% operational cost increase would severely degrade Sri Lanka's competitive pricing in the South Asian tourism market. Tourism remains a primary conduit for foreign exchange (Forex) liquidity.[1] Consequently, small and medium-scale tourism operators officially classified under the Hotel

(H-1 and H-2) categories were granted a complete exemption from the May 11, 2026, tariff hike, provided their consumption does not exceed excessively anomalous upper-bound thresholds.[8]

This creates a distinct dual-tier reality for tourist accommodations. A villa that has circumvented formal registration and continues to operate on a standard commercial GP-2 meter will suffer a debilitating financial penalty compared to a legally compliant, Sri Lanka Tourism Development Authority (SLTDA) registered villa operating on an H-2 meter. This regulatory reality must form the baseline assumption for any financial modeling executed by an Energy Management System deployed in this sector.

Navigating CEB Tariff Classifications: The Imperative of the H-2 Standard

For the dynamic tariff calculation module of the proposed IoT EMS to function accurately, it must be programmed to differentiate between the structural mechanisms of the CEB's tariff classifications.[1] Understanding the path to securing these rates, and the internal mechanisms of the rates themselves, is fundamental to the system's economic efficacy.

The SLTDA Registration Barrier

Under CEB guidelines, the Hotel (H) category is not granted by default; it is explicitly reserved for premises formally approved and regulated by the Sri Lanka Tourism Development Authority (SLTDA).[13] The barrier to entry for this classification involves a rigorous compliance and registration protocol. Accommodations must submit extensive documentation, including renewed Environmental Protection Licenses (EPL), public liability insurance policies, comprehensive fire protection certificates, water quality microbiology test reports, and formal business trade licenses from local authorities.[14]

Only upon fulfilling these stringent criteria and undergoing physical compliance inspections does a property earn its classification, allowing the CEB to transition the property's meter from a General Purpose designation to a highly coveted Hotel designation.[16] If a property owner fails to maintain this registration, they are legally classified as a standard commercial entity and subjected to the heavily penalized GP-2 rates.

The H-1 vs. H-2 Distinction: VDMC versus Native Time-of-Use

Once an accommodation secures the Hotel classification, its specific tariff sub-category is determined by its electrical infrastructure's contract demand. The CEB fundamentally bifurcates this category:

1. **Hotel Category H-1:** This classification applies to supplies at each individual point of supply delivered and metered at 400/230 Volt nominal where the contract demand is less than or equal to 42 kVA.[13] Crucially, the base H-1 category historically operates on a Volume Differentiated Monthly Consumption (VDMC) framework.[18] The VDMC is essentially a flat-rate block structure that penalizes total aggregate volume rather than

penalizing specific hours of consumption. Because it lacks a native temporal dimension, it is highly rigid and poorly suited for dynamic real-time arbitrage.

2. **Hotel Category H-2:** This classification applies to supplies where the contract demand explicitly exceeds the 42 kVA threshold.[13] Unlike the flat-rate structure of H-1, the H-2 category operates on a native, continuous Time-of-Use (ToU) structure.[13] This structure severely penalizes consumption during grid peak hours while heavily discounting nocturnal baseloads.

Because a tourist villa or guesthouse legally qualifies for the Hotel Category, and because the entire foundational premise of the IoT EMS is to maximize financial arbitrage across moving time windows [1], the system's mathematical engine must be explicitly optimized for the H-2 (Hotel Rate 2) ToU standards. H-2 is the correct baseline when referencing specific ToU algorithmic structures.

Time-of-Use (ToU) Load Profiling and Hospitality Behavioral Dynamics

The architectural design of the proposed IoT Energy Management System relies on intercepting the specific behavioral patterns of tourist guests and mapping those patterns against the temporal constraints of the utility provider.[1]

The Specific ToU Temporal Blocks

The Ceylon Electricity Board enforces three strict, immalleable Time-of-Use windows across all applicable ToU categories. These specific temporal boundaries are the primary trigger conditions that must be programmed into the system's backend logic to execute real-time multiplier shifts.[13] The exact blocks are defined as:

ToU Window Designation	Temporal Boundary	Total Duration	Operational Context
Day	05:30 to 18:30	13 Hours	Standard daylight operations; moderate grid load.
Peak	18:30 to 22:30	4 Hours	Maximum grid stress; highest financial penalty.

Off-Peak	22:30 to 05:30	7 Hours	Nocturnal baseload; lowest financial penalty.

The Peak Window Vulnerability and Behavioral Wastage

In tropical environments like Sri Lanka, the peak utility window perfectly intersects with the peak physiological and behavioral demands of tourists. Returning to their accommodations from day excursions typically between 17:00 and 19:00, guests routinely engage in sudden, high-energy activities. This includes showering (activating 2000W to 3000W geysers and water heaters), turning on comprehensive room lighting arrays, and engaging HVAC systems. These HVAC units are frequently set to minimum permissible temperatures (e.g., 16°C) and maximum fan speeds to rapidly displace the substantial solar heat gain the room absorbed throughout the day.

The critical point of failure occurs shortly thereafter. If guests subsequently leave their rooms for dinner or evening entertainment during this window—a highly common hospitality pattern—the HVAC systems and geysers remain running continuously at maximum operational load.[1] Because this unmonitored runtime occurs precisely during the 18:30 to 22:30 peak window, the energy wastage is disproportionately expensive.

Relying on guests to manually switch off their 1500W air conditioning units precisely at 18:30 is an operational impossibility. Similarly, traditional key-card slots are easily circumvented by guests leaving secondary cards or thick paper in the slot to keep the room cool while they dine.[1] Consequently, an automated intervention is mathematically necessitated to preserve the accommodation's fiscal health.

Exhaustive Tariff Matrix: May 2026 Post-Revision Financial Baselines

To fulfill the specific functional requirements outlined in the project's Gap Analysis—namely FR-COST-01 (System shall store configurable CEB tariff rates) and FR-COST-03 (System shall support time-of-use tariff windows) [1]—the exact numerical parameters of the May 2026 tariff revisions must be established. The baseline must clearly contrast the protected H-2 category against the penalized GP-2 category to highlight the vital necessity of SLTDA registration and accurate system optimization.

Hotel Rate 2 (H-2): The Recommended Operational Baseline

Since small and medium-scale accommodations under the H-1 and H-2 classifications are strategically protected from the May 2026 18% hike [8], the H-2 rates remain steady at their established 2025 baselines. This provides a predictable, albeit still highly stratified, cost environment for the EMS to optimize against.

The precise numerical breakdown for the H-2 ToU structure is as follows:

Tariff Component	ToU Window	Applicable Rate (LKR)
Off-Peak Energy Charge	22:30 to 05:30	12.00 per kWh
Day Energy Charge	05:30 to 18:30	15.00 per kWh
Peak Energy Charge	18:30 to 22:30	28.00 per kWh
Monthly Fixed Charge	System Wide	5,000.00 LKR
Maximum Demand Charge	System Wide	1,400.00 per kVA

Even within this protected category, the ToU penalty is severe. The Peak rate of 28.00 LKR is 2.33 times more expensive than the Off-Peak rate of 12.00 LKR. This steep economic gradient is the primary lever the dynamic tariff algorithm will manipulate to generate ROI.

General Purpose Rate 2 (GP-2): The Unregistered Penalty Baseline

For comparison, it is vital to programmatic analysis to understand the consequences of a property operating without official SLTDA Hotel registration. Such properties default to the General Purpose Rate 2 (GP-2) category. This category absorbed the full brunt of the May 11, 2026, tariff revision, facing a direct 18% scalar increase across both its energy blocks and fixed infrastructure charges.[8]

The mathematically adjusted rates for the GP-2 category under the new PUCSL directive are:

Tariff Component	Pre-Hike Baseline	May 2026 18% Adjustment	Final GP-2 Rate (LKR)

Off-Peak Energy Charge	31.00 LKR/kWh	31.00 + 18%	36.58 per kWh
Day Energy Charge	41.00 LKR/kWh	41.00 + 18%	48.38 per kWh
Peak Energy Charge	47.00 LKR/kWh	47.00 + 18%	55.46 per kWh
Monthly Fixed Charge	5,000.00 LKR	5,000.00 + 18%	5,900.00 LKR
Maximum Demand Charge	1,500.00 LKR/kVA	1,500.00 + 18%	1,770.00 per kVA

Financial Delta Analysis and Third-Order Implications

The disparity between these two categories highlights the catastrophic financial risk of operating a tourist accommodation on a standard commercial meter in post-May 2026 Sri Lanka. During the Peak window (18:30 – 22:30), an unregistered villa pays an exorbitant **55.46 LKR per kWh**. This is nearly double the **28.00 LKR per kWh** paid by an officially registered H-2 counterpart.

Simulated Load Case Study:

Consider a behavioral waste scenario where a guest leaves a room vacant from 19:00 to 22:00 (3 hours squarely within the Peak window). The room contains a 1.5 kW inverter air conditioner, a 2.0 kW rapid water heater, and 0.2 kW of assorted lighting and peripheral loads. The total concurrent load is 3.7 kW.

- **Total Energy Hemorrhaged:** 3.7 kW × 3 hours = 11.1 kWh.
- **Cost Incurred under GP-2:** 11.1 kWh × 55.46 LKR = 615.60 LKR per single event.
- **Cost Incurred under H-2:** 11.1 kWh × 28.00 LKR = 310.80 LKR per single event.

If an SME property consisting of 10 rooms experiences this exact behavioral pattern just 15 days a month, the peak-hour leakage equates to 46,620 LKR monthly under the optimized H-2 rate, and an astronomical 92,340 LKR monthly under the GP-2 rate. Over a fiscal year, this single behavioral inefficiency results in over half a million LKR in lost revenue for a protected hotel, and over a million LKR for an unprotected one.

The IoT EMS targets this exact vulnerability. By detecting absolute vacancy, the system instantly severs the relay connections to the heavy loads, executing automated demand-side load shedding exactly when the grid pricing is most punitive.[1]

System Architecture: IoT Integration and Edge Computing

To reliably execute load shedding without compromising guest comfort or generating negative reviews, the proposed system employs a rigorous Constructive Research Method, operating via an Agile Prototyping Model.[1] The architecture is designed to overcome the prohibitive costs of enterprise Building Management Systems (BMS) while matching their operational efficacy. The system spans three distinct interconnected layers: the Perception Layer, the Network Layer, and the Application Layer.[1]

The Perception Layer: Edge Node Processing

The physical interface between the digital system and the real world occurs at the Perception Layer. This layer relies heavily on edge computing nodes powered by ESP32 microcontrollers.[1] The ESP32 is a low-cost, low-power system-on-a-chip (SoC) equipped with integrated Wi-Fi and dual-core processing capabilities, making it ideal for distributed IoT deployments in hospitality settings.

The ESP32 interfaces directly with a suite of environmental sensors and acts as the local logic controller. It triggers high-amperage relays that physically cut the alternating current (AC) to non-essential appliances like air conditioners and geysers when rooms are definitively unoccupied.[1]

A critical component of the Perception Layer is safety and isolation. The prototype involves controlling lethal 230V mains electricity. To ensure absolute electrical safety and compliance with international standards, the 5V DC control circuits originating from the ESP32 are strictly optically isolated from the high-voltage load using high-grade optocouplers.[1] This guarantees that power surges or relay malfunctions cannot feed dangerous voltage back into the delicate low-voltage microprocessor logic boards. Furthermore, all hardware is mandated to be enclosed within non-conductive, fire-retardant custom casings.[1]

The Network Layer: Transitioning to Realtime Database Synchronization

Historically, lightweight IoT telemetry has relied heavily on the MQTT (Message Queuing Telemetry Transport) protocol, a publish-subscribe network protocol designed for constrained devices and low-bandwidth, high-latency networks.[1] Indeed, the original project proposal explicitly mandated an MQTT architecture.[1]

However, a comprehensive gap analysis of the evolving system architecture reveals a strategic pivot away from traditional MQTT brokers in favor of a direct synchronization model utilizing the **Firebase Realtime Database (RTDB)**.[1] This shift drastically simplifies the network topology by removing the intermediate MQTT broker layer. Instead of publishing to a broker which is then parsed by a backend server, the ESP32 edge nodes use Firebase REST APIs or WebSocket connections to push sensor readings directly into the cloud database in real-time.[1]

This direct-sync architecture reduces latency between the edge perception nodes and the Application Layer, allowing for millisecond-level updates to power status, vacancy triggers, and cost accumulation metrics.[1] The gap analysis explicitly identifies the need to phase out simulated MQTT hooks (e.g., useMQTTSimulation.ts) in the frontend codebase and replace them with robust Firebase listener integrations.[1]

The Application Layer: The MERN Stack and Dashboard Visualization

The final layer is the Application Layer, which serves as the primary interface for the property owner. Developed using the MERN stack—comprising MongoDB (for potential long-term data warehousing), Express.js, React (for the user interface), and Node.js (for backend API routing)—the system provides a centralized, secure web-based dashboard.[1]

The React-based frontend aggregates the real-time telemetry streaming from Firebase. It allows non-technical hospitality managers to remotely monitor the live occupancy status of every room, execute manual override controls if necessary (e.g., pre-cooling a room for a VIP guest before arrival), and visualize complex analytics regarding energy consumption patterns and financial expenditures.[1]

The Hybrid Occupancy Detection Algorithm: Solving the "False Negative" Paradigm

The entire premise of automated load shedding rests upon the system's ability to definitively ascertain human presence. Traditional legacy systems in SME hospitality rely heavily on standalone Passive Infrared (PIR) sensors. This reliance is deeply flawed.[1]

Standard PIR sensors operate by measuring changes in ambient infrared radiation across segmented spatial grids. They detect moving thermal signatures. Consequently, they are highly prone to "False Negatives".[1] If a tourist guest is physically present in the room but largely stationary—such as reading a book in a chair, working quietly on a laptop, or sleeping in bed—the PIR sensor will register a lack of dynamic movement. A simplistic system will interpret this stillness as vacancy and disastrously plunge the room into darkness while terminating the air conditioning. This generates extreme hostility from guests, leading to highly damaging negative online reviews that can cripple an SME's reputation.

To engineer around this fundamental hardware limitation, the IoT EMS discards single-sensor dependency in favor of a proprietary **Hybrid Occupancy Detection Algorithm** based on rigorous multi-sensor data fusion.[1]

The system integrates three primary datastreams:

1. **PIR Motion Sensors:** To detect active thermal movement within the room's core volume.[1]
2. **Magnetic Door Contacts (Reed Switches):** Installed on the main entry doors (and potentially balcony doors) to log the absolute physical state changes of the room's primary entry and exit vectors.[1]
3. **Current Transformers (CT Sensors):** To analyze the distinct electrical noise profiles and baseline loads of active human intervention versus passive standby states.[1]

State Machine Logic: Creating Context-Aware "Room States"

Rather than generating a simplistic binary Boolean output (1 = Occupied, 0 = Vacant), the ESP32 microcontroller at the edge computes a sophisticated state machine. The algorithm evaluates the chronological sequence of sensor events to categorize the room into fluid, context-aware behavioral states [1]:

- **State 1: Occupied-Active:** This state is triggered when the magnetic door contact registers an "Open" followed by a "Close" event, and the PIR sensor subsequently detects continuous or varied thermal movement. The system deduces a guest has entered and is moving about. It maintains all high-voltage power relays in the CLOSED (On) position, ensuring full amenity availability.[1]
- **State 2: Occupied-Sleeping (or Stationary):** This is the critical innovation of the hybrid logic. This state is initiated when the system previously registered 'Occupied-Active', but PIR movement ceases entirely for a sustained period. Crucially, because the magnetic door contact *has not registered a subsequent exit event* (an opening and closing of the door), the algorithmic logic deduces that the human entity is still within the physical bounds of the room but is simply stationary. The system purposefully blocks the relays from opening, prioritizing continuous HVAC and lighting comfort over aggressive energy culling, entirely eliminating the "False Negative" sleeping guest problem.[1]
- **State 3: Vacant:** This state is definitively triggered only when a distinct door open/close exit event is logged by the magnetic contact, followed immediately by a sustained cessation of PIR motion events within the room.[1] Upon entering State 3, the ESP32 initiates a programmable timeout sequence (e.g., a 5 to 10-minute buffer to prevent false triggers if a guest merely steps into the corridor to speak to housekeeping). Once the timeout buffer expires without further PIR interference, the ESP32 logic opens the high-load relays, neutralizing the HVAC and geyser circuits and halting energy hemorrhage.[1]

This context-aware adaptation ensures the system solves the energy crisis inherent to SME tourism without generating friction that could damage the core hospitality product.[1]

Data Model Evolution: The Firebase RTDB JSON

Schema

As identified in the project's Gap Analysis, fulfilling the critical functional requirements surrounding cost tracking demands a highly structured data storage strategy. Specifically, FR-COST-01 dictates that the system must store configurable CEB tariff rates, while FR-COST-03 dictates the system must support specific time-of-use tariff windows.[1]

A major security and operational flaw in early prototyping involves hardcoding these tariff variables directly into the React frontend source code or the ESP32 firmware.[1] Tariffs, as painfully demonstrated by the sudden May 9, 2026, PUCSL intervention, are inherently volatile macroeconomic instruments. They require a dynamic architecture that permits remote, over-the-air parameter updates via the dashboard's settings menu (FR-COST-07) without necessitating complex firmware flashes or software redeployments.[1]

To support this, the Firebase schema must be aggressively refactored. The current flat topology (tracking generic overarching nodes like sensors or devices) must be replaced with a nested, multi-tenant hierarchical tree centered around specific properties and rooms.[1]

The Optimized Firebase tariffs JSON Schema

To guarantee that the dynamic calculation module accurately reflects real-world costs and avoids hardcoding penalties, the exact May 2026 H-2 baseline rates and precise ToU time boundaries must be mapped structurally to the specific property's JSON node within the Firebase Realtime Database.

The following JSON structure explicitly maps the recommended H-2 ToU parameters, ensuring the system can reliably pull the exact active multipliers:

JSON

```json
{
 "tariffs": {
  "category": "H-2",
  "currency": "LKR",
  "fixedCharge": 5000,
  "offPeak": {
    "start": "22:30",
    "end": "05:30",
    "rate": 12.00
  },
  "day": {
    "start": "05:30",
```

```json
    "end": "18:30",
    "rate": 15.00
   },
   "peak": {
    "start": "18:30",
    "end": "22:30",
    "rate": 28.00
   }
 }
}
```

This strict hierarchical relationship solves multiple software engineering challenges simultaneously. When the frontend React dashboard renders the Analytics.tsx view, it effortlessly parses the tariffs object bound to the specific property ID. Furthermore, Phase 7 of the system's development roadmap addresses critical security gaps (FR-SEC-01 to FR-SEC-04) by imposing robust Firebase Authentication mechanisms and stringent database access rules.[1] By nesting the structure appropriately, Firebase Security Rules can be written to ensure that a property owner's authenticated User ID (UID) only possesses read/write privileges over their specific property tree, preventing data leakage or malicious tampering across multiple registered tourist villas.

The Dynamic Tariff Calculation Algorithm: Mathematical and Programmatic Logic

With the perceptual edge hardware reliably logging occupancy states, and the Firebase RTDB securely managing the volatile tariff constraints, the core software requirement of the project lies in the mathematical translation of raw physics into actionable economic metrics. Research Objective OBJ-03 explicitly demands the implementation of an algorithm that calculates estimated cost savings based strictly on active CEB tariff rates.[1]

Currently identified as a "Major Gap" in the system's development, the existing application relies on static, hardcoded UI components to display "cost savings" that lack actual mathematical validity.[1] Moving to a fully automated production environment requires engineering a Dynamic Tariff Calculation Algorithm to handle the non-linear, time-dependent realities of the ToU metering structure.[1]

The Mathematical Foundation of the ToU Algorithm

A rudimentary energy calculation algorithm utilizes a simplistic, monolithic formula:

$ \text{Total Cost} = \text{Total Consumed kWh} \times \text{Average Tariff Rate} $

However, this fundamentally fails to exploit the CEB's ToU structure, completely masking the devastating financial drain of peak-hour wastage and violating the core intent of the system.[1]

The proposed advanced calculation logic dictates that total energy costs must be aggressively segregated and aggregated as the sum product of isolated consumption events occurring exclusively within their predefined temporal windows.[1]

Let E represent the total energy cost over a given continuous 24-hour cycle. The comprehensive algorithmic logic is modeled as:

$$E = \sum(kWh_{op} \cdot R_{op}) + \sum(kWh_d \cdot R_d) + \sum(kWh_p \cdot R_p)$$

Where:

- kWh_{op}, kWh_d, and kWh_p represent the precise kilowatt-hours consumed by the room during the Off-Peak, Day, and Peak windows, respectively.
- R_{op}, R_d, and R_p are the active H-2 tariff multipliers (12.00, 15.00, and 28.00 LKR) queried dynamically in real-time from the Firebase tariffs node.[1]

Integrating Server Timestamps and Edge Case Mitigation

To execute this mathematical formula reliably, the system must synchronize the physical power readings with absolute temporal precision. The power monitoring module (e.g., a PZEM-004T) interfaced with the ESP32 continuously calculates accumulated energy (kWh).

The algorithm requires robust timestamping to govern the multiplier selection. As the ESP32 pushes a data payload to the Firebase RTDB, the backend logic must intercept this payload, parse the server's absolute current timestamp (typically utilizing the UNIX epoch standard), and evaluate it against the strictly defined ToU boundaries defined in the JSON schema (e.g., assessing if the timestamp falls exactly between the strings "18:30" and "22:30").

Edge Case Mitigation: Cross-Boundary Calculation Slicing

A critical complexity in the programming algorithm arises during cross-boundary events. The system cannot simply assign an entire usage session to a single tariff block if the session spans a temporal boundary.

For example, assume a guest vacates the room at 18:20. The ESP32 is programmed with a 15-minute relay timeout buffer to definitively ensure vacancy and prevent a false trigger. Consequently, the HVAC will remain running and drawing power until 18:35. This 15-minute block of power consumption crosses the boundary between the Day rate (15.00 LKR) and the Peak rate (28.00 LKR) precisely at 18:30.

The algorithm must be engineered to computationally slice the energy integral at the boundary threshold. It must assign the first 10 minutes of consumption (from 18:20 to 18:30) to the kWh_d day variable, and the final 5 minutes of consumption (from 18:30 to 18:35) to the

kWh_p peak variable. Without this high-resolution fractional precision, the system risks either under-reporting or over-reporting the financial data, fatally undermining the integrity of the Return on Investment (ROI) metric presented to the business owner.

Real-Time ROI Quantification and "Money Saved" Metrics

Beyond merely computing the incurred utility liability, the true psychological and commercial value of the algorithm lies in dynamically computing the *avoided cost*.[1] Standard automation systems display generic historical energy usage charts. This system innovates by providing Real-Time ROI Quantification.[1]

When the Hybrid Occupancy Detection logic triggers State 3 (Vacant) and commands the relays to open, terminating the flow of AC power to the heavy appliances, the system initiates a virtual logic sequence. If the room was actively consuming 2.5 kW at the exact moment of relay disconnection, the algorithm initiates a virtual counter based on that theoretical load.

Let S equal the total LKR saved during a vacancy event of duration t (measured in hours):

$$S = \int_0^t (P_{theoretical} \times R_{active})dt$$

Where $P_{theoretical}$ is the projected continuous load of the appliances had they remained engaged by an absent guest, and R_{active} is the dynamically queried CEB tariff rate applicable at that exact moment in time based on the ToU windows.

By routing this continuous, calculus-driven algorithmic output to the MERN/React frontend, the dashboard fulfills requirement FR-COST-05 by projecting a live, ticking counter of "Money Saved (LKR)".[1] This constant, visible financial feedback loop provides absolute transparency and is instrumental in justifying the capital expenditure of the IoT EMS for the SME owner.[1]

Empirical Validation Framework and ROI Assessment Strategies

To rigorously validate the efficacy of the physical hardware edge nodes, the stability of the Firebase RTDB synchronization, and the mathematical accuracy of the Dynamic Tariff Calculation Algorithm, the project must deploy a highly structured, objective empirical testing methodology.[1]

The research utilizes a "Pre-test/Post-test" Comparative Energy Analysis, specifically designed to eliminate external variables and provide a clean assessment of algorithmic efficiency.[1]

The 48-Hour Controlled Validation Matrix

The evaluation matrix operates over a 48-hour continuous cycle to benchmark the system against a defined null hypothesis—namely, the assumption that the IoT EMS provides negligible financial benefit over manual operations.[1]

Phase 1: The Control Setup (Baseline Generation) For the initial 24-hour period, an isolated test environment—a simulated or actual hotel room containing standard HVAC units (e.g., 12000 BTU inverter AC) and typical lighting arrays—is subjected to unmonitored, manual operation.[1] Crucially, the IoT EMS is installed but placed in a passive "listening" mode; it logs all sensor data, movement patterns, and timestamps but is programmatically restricted from triggering the relays to shed loads.

During this phase, human actors introduce simulated "behavioral waste" by vacating the room for hours at a time (specifically targeting the 18:30 to 22:30 window) while deliberately leaving all appliances active. The system logs the total baseline consumption ($kWh_{baseline}$) and calculates the theoretical baseline cost in LKR using the dynamic ToU algorithm.[1]

Phase 2: The Experimental Setup (Automated Mitigation) For the subsequent 24-hour period, the IoT EMS is fully armed and transitioned to active mode. The exact same sequence of human occupancy and behavioral waste is simulated by the actors.[1] However, this time, the system's Hybrid Occupancy Detection logic actively intercepts the vacancy states. Upon confirming a State 3 (Vacant) condition and passing the timeout buffer, the system physically sheds the HVAC and lighting loads. The system logs the total automated consumption ($kWh_{automated}$) and calculates the mitigated cost in LKR.

Assessment Metrics and the 20% Success Threshold

The definitive success of the system relies on proving FR-EVAL-04 (Recording total LKR costs for both test periods) and executing the final comparative analysis.[1] The metric for absolute success is defined mathematically as the Energy Reduction Percentage:

$$\text{Energy Reduction} = \left(\frac{kWh_{baseline} - kWh_{automated}}{kWh_{baseline}} \right) \times 100$$

The project's architectural mandate aims to achieve an aggregate reduction in energy wastage—specifically targeting unoccupied runtime—of at least **20%** compared to the unmanaged baseline.[1]

Given that the testing matrix intentionally weighs behavior heavily during the 18:30 to 22:30 peak window, mitigating unoccupied runtime precisely during this 4-hour block guarantees a severely disproportionate reduction in the *financial cost* relative to the raw *energy reduction*. Because the algorithm dynamically pulls the 28.00 LKR peak multiplier rather than the 12.00 LKR off-peak multiplier, shedding 20% of the raw energy during the peak window results in an exponentially higher financial savings yield than shedding 20% of the energy during the nocturnal block. This proves the unparalleled value of integrating context-aware ToU parameters directly into the IoT logic.

Strategic Industry Implications and Concluding Outlook

Sri Lanka's SME tourism sector operates within an unforgiving macroeconomic crucible. The aggressive May 9, 2026, PUCSL tariff revisions—culminating in a sweeping 18% hike for General Purpose commercial consumers—underscore the highly volatile reality of a thermal-heavy national grid generation strategy facing imported fossil-fuel inflation.[2]

While the strategic regulatory exemption of the H-1 and H-2 Hotel categories shields legally compliant, SLTDA-registered operators from the immediate devastation of the 18% shock [8], it absolutely does not eliminate the severe operational penalties baked into the native Time-of-Use architecture of the H-2 rate itself. Peak hour electricity remains exorbitantly expensive at 28.00 LKR per unit, and unmitigated guest behavioral waste continues to pose an existential threat to SME profit margins.

The "Smart IoT-Based Energy Management System" provides a highly localized, deeply contextualized technological remedy to this crisis. By transitioning away from easily spoofed key-card switches and flawed single-sensor PIR networks, the system leverages an advanced multi-sensor Hybrid Occupancy Detection logic to navigate the complex, sensitive boundary between aggressive demand-side load-shedding and uncompromised guest comfort.[1] Furthermore, by abandoning restrictive flat-file data structures in favor of a scalable, property-nested Firebase Realtime Database schema, the system supports highly secure, multi-tenant deployment architectures capable of dynamic, over-the-air tariff adjustments.[1]

Ultimately, the true transformative value of the system lies in its algorithmic sophistication. By computationally slicing real-time physical power metrics against the exact temporal boundaries of the CEB's ToU windows, the Dynamic Tariff Calculation Algorithm delivers granular, highly accurate Return on Investment data.[1] This empowers non-technical business owners with unprecedented real-time financial transparency. It effectively bridges the technological void between prohibitively expensive commercial Building Management Systems and the urgent, immediate survival requirements of Sri Lanka's vital SME tourism operators. The continuous automated mitigation of peak-hour loads not only ensures commercial fiscal resilience for individual villas but actively contributes to aggregate national grid stability, reinforcing Sri Lanka's global positioning as a forward-thinking, sustainable tourism destination.

Works cited

1. PROPOSAL_GAP_ANALYSIS_AND_FUNCTIONAL_REQUIREMENTS.md
2. Revised electricity tariffs to take effect from Monday; Govt subsidy in place until September., accessed May 10, 2026, https://dailynews.lk/2026/05/10/local/992217/revised-electricity-tariffs-to-take-effect-from-monday-govt-subsidy-in-place-until-september/

3. PUCSL approves 18% increase in electricity tariffs - Newsfirst.lk, accessed May 10, 2026, https://www.newsfirst.lk/2026/05/09/pucsl-approves-18-increase-in-electricity-tariffs
4. Paying Rs. 17,000 for Electricity? Your bill could soon hit Rs. 20,000 - Newswire, accessed May 10, 2026, https://www.newswire.lk/2026/05/09/paying-rs-17000-for-electricity-your-bill-could-soon-hit-rs-20000/
5. Revised electricity tariffs effective from Monday; government subsidy until September, accessed May 10, 2026, https://www.newswire.lk/2026/05/10/revised-electricity-tariffs-effective-from-monday-government-subsidy-until-september/
6. Electricity tariffs increased by 18% for 5% of consumers | The Morning - Themorning.lk, accessed May 10, 2026, https://www.themorning.lk/articles/HNpCda1GFxuaxSHoOani
7. Electricity tariffs to increase by 18% for domestic users exceeding 180 units - DailyNews, accessed May 10, 2026, https://dailynews.lk/2026/05/09/breaking-news/992153/electricity-tariffs-to-increase-by-18-for-domestic-users-exceeding-180-units/
8. Electricity tariffs to rise by 18% for high-consumption users - Breaking News - Daily Mirror, accessed May 10, 2026, https://www.dailymirror.lk/breaking-news/Electricity-tariffs-to-rise-by-18-for-high-consumption-users/108-339876
9. accessed May 10, 2026, https://www.dailymirror.lk/print/breaking-news/Electricity-tariffs-to-rise-by-18-for-high-consumption-users/108-339876#:~:text=Colombo%2C%20May%2009%20(Daily%20Mirror,monthly%20consumption%20exceeds%20180%20units.
10. CEB requests 13.56% electricity tariff hike for 2026 second quarter - Ada Derana, accessed May 10, 2026, http://www.adaderana.lk/news/118434/ceb-requests-1356-electricity-tariff-hike-for-2026-second-quarter
11. CEB proposes electricity tariff hike for second quarter of 2026 - MidPoint.lk, accessed May 10, 2026, https://midpoint.lk/ceb-proposes-electricity-tariff-hike-for-second-quarter-of-2026/
12. Electricity Prices Increased in Sri Lanka - Newswire, accessed May 10, 2026, https://www.newswire.lk/2026/05/09/electricity-prices-increased-in-sri-lanka/
13. Hotels - Tariff - PUCSL, accessed May 10, 2026, https://www.pucsl.gov.lk/electricity/tariff/hotels/
14. RegistrationProcess – Classified Hotel - SLTDA, accessed May 10, 2026, https://www.sltda.gov.lk/storage/common_media/Classified%20Hotels-up.pdf
15. Registration – Classified Hotel Process - Sri Lanka Tourism Development Authority, accessed May 10, 2026, https://sltda.gov.lk/storage/common_media/Classified%20Hotels%20-%20F796688929.pdf

16. Registration Process – Tourist Hotel, accessed May 10, 2026, https://www.sltda.gov.lk/storage/common_media/Tourist%20Hotel%20Process%20&%20Document%20Requirement.pdf
17. CEB Tariff Categories Overview | PDF | Technology & Engineering - Scribd, accessed May 10, 2026, https://www.scribd.com/document/382759366/CEB-Tariff-Categories
18. Ceylon Electricity Tariff Update 2025 | PDF - Scribd, accessed May 10, 2026, https://www.scribd.com/document/954361161/Second-Tariff-Revision-2025-E-Copy-3
19. Electricity Tariff Revision 2022 - CEB, accessed May 10, 2026, https://ceb.lk/front_img/1660192838Electricity-Tariff_Revision-2022.pdf
20. Electricity - Tariff - Industrial - PUCSL, accessed May 10, 2026, https://www.pucsl.gov.lk/electricity/tariff/industrial/